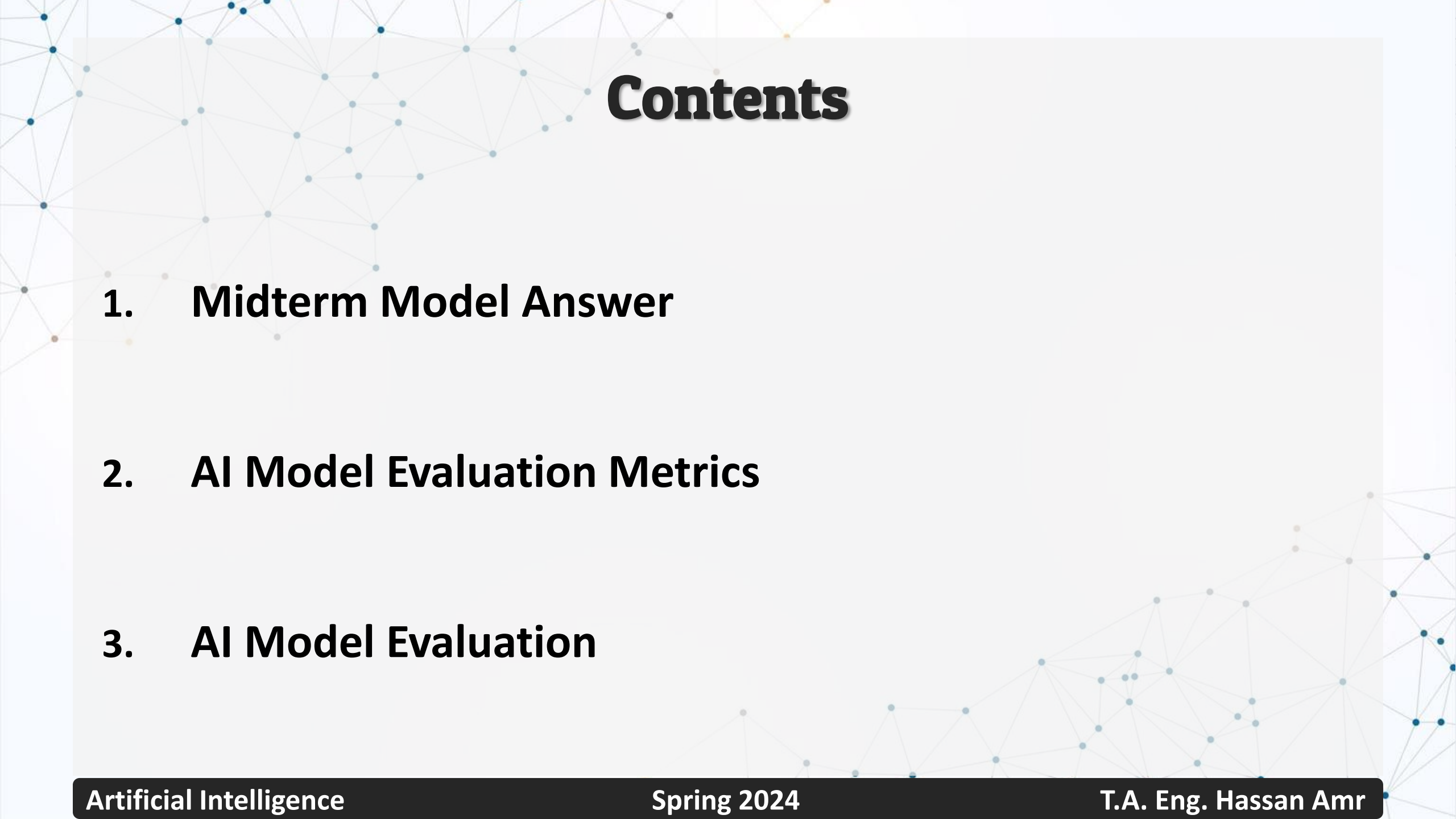


Artificial Intelligence (Spring 2024)

AI Week 11

**Midterm Model Answer,
AI Model Evaluation Metrics,
AI Model Evaluation**



Contents

- 1. Midterm Model Answer**
- 2. AI Model Evaluation Metrics**
- 3. AI Model Evaluation**

Midterm Model Answer

Question 1: Building autonomous robots that assist people in everyday life has been a long-standing goal of research in AI and robotics. As a case study, consider a tour guide robot called Promobot, working in museums and exhibits as shown in Fig.1

(a) Is this robot could be considered as a line follower robot (LFR)?

Yes, by following a trajectory on ground to navigate inside the museum.

Midterm Model Answer

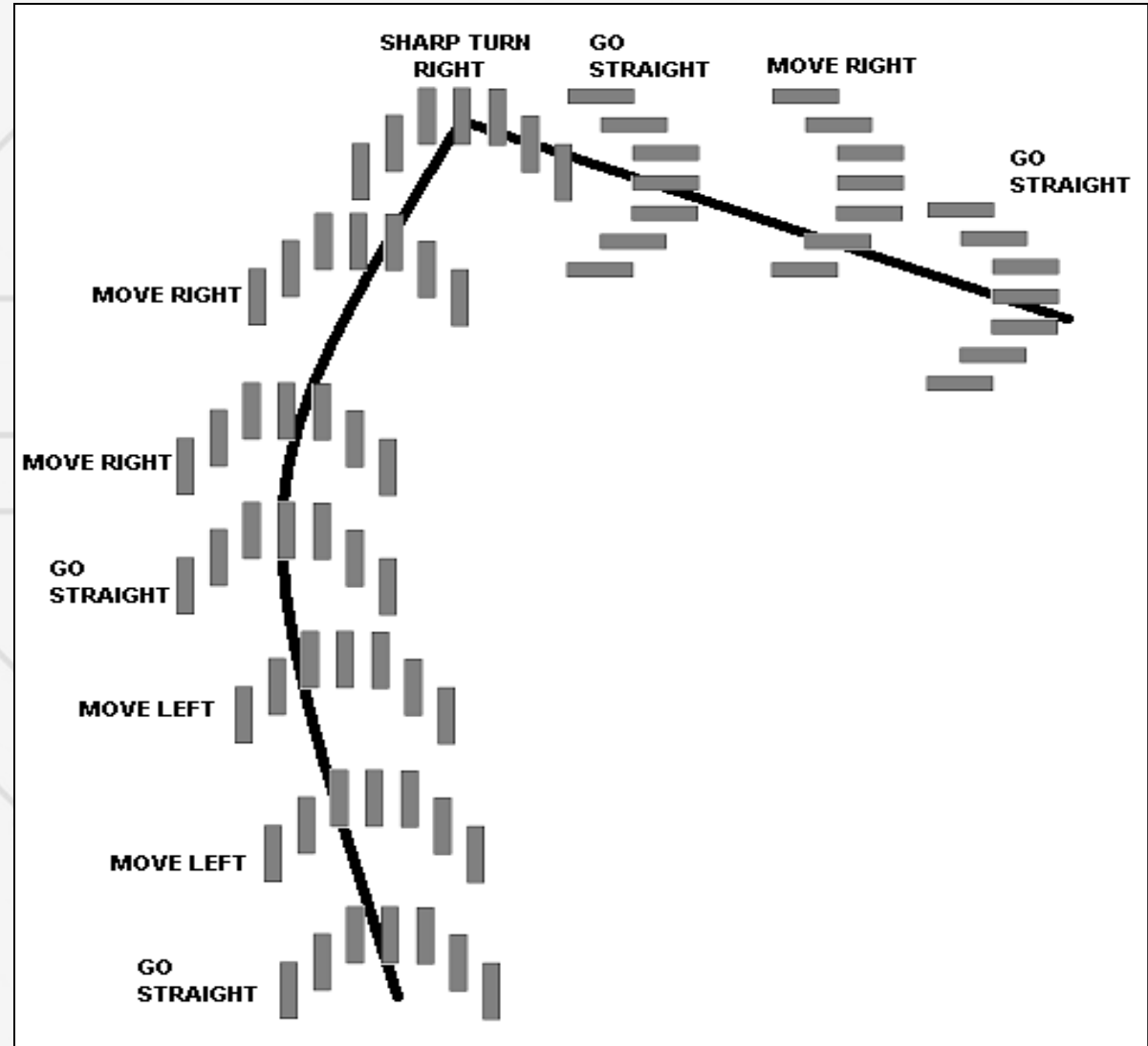
(b) What are the operational requirements of this robot?

LFR REQUIREMENTS

1. The robot must be capable of **following a line**.
2. It should be capable of taking **various degrees of turns**
3. It must be prepared of a situation that it runs into a territory **which has no line to follow**.
4. The robot must also be capable of following a **line even if it has breaks**.
5. The robot must be insensitive to environmental factors such as **lighting and noise**.
6. It must allow calibration of the **line's darkness threshold**.
7. **Scalability** must be a primary concern in the design.
8. The **color** of the line **must not be a factor** as long as it is darker than the surroundings.

Midterm Model Answer

(C) Draw the trajectory on earth for the Promobot.

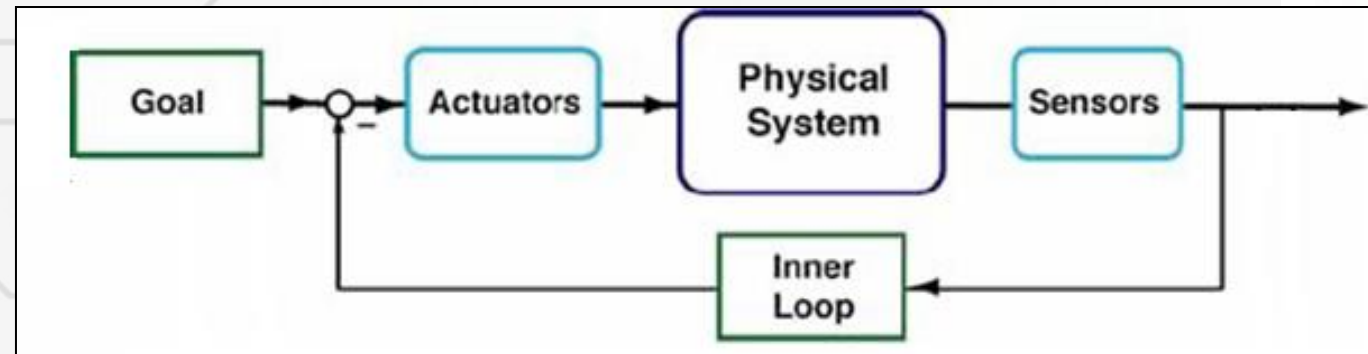


Midterm Model Answer

(d) What subsystems make up such a robot?

This robot consists of:

- **Action**
 - **Stationary base**
 - **Mobile**
- **Sensors**
- **Control**
- **Power supply**



Midterm Model Answer

(e) What are the types of essential sensors of this robot?

- **Optical**
 - o **Laser / radar**
 - o **3D**
 - o **Color spectrum**
- **Pressure**
- **Temperature**
- **Motion & Accelerometer**

Midterm Model Answer

(f) How does this robot should move?

Using Legs

(g) How do you measure the performance of such a robot?

- **Speed and acceleration**
- **Resolution**
- **Working volume**
- **Accuracy**
- **Cost**

Midterm Model Answer

Question 2:

In 2050, we can expect to see great advancements in technology, particularly in the areas of artificial intelligence (AI) and automation. These technologies will revolutionize various industries and significantly impact our daily lives.

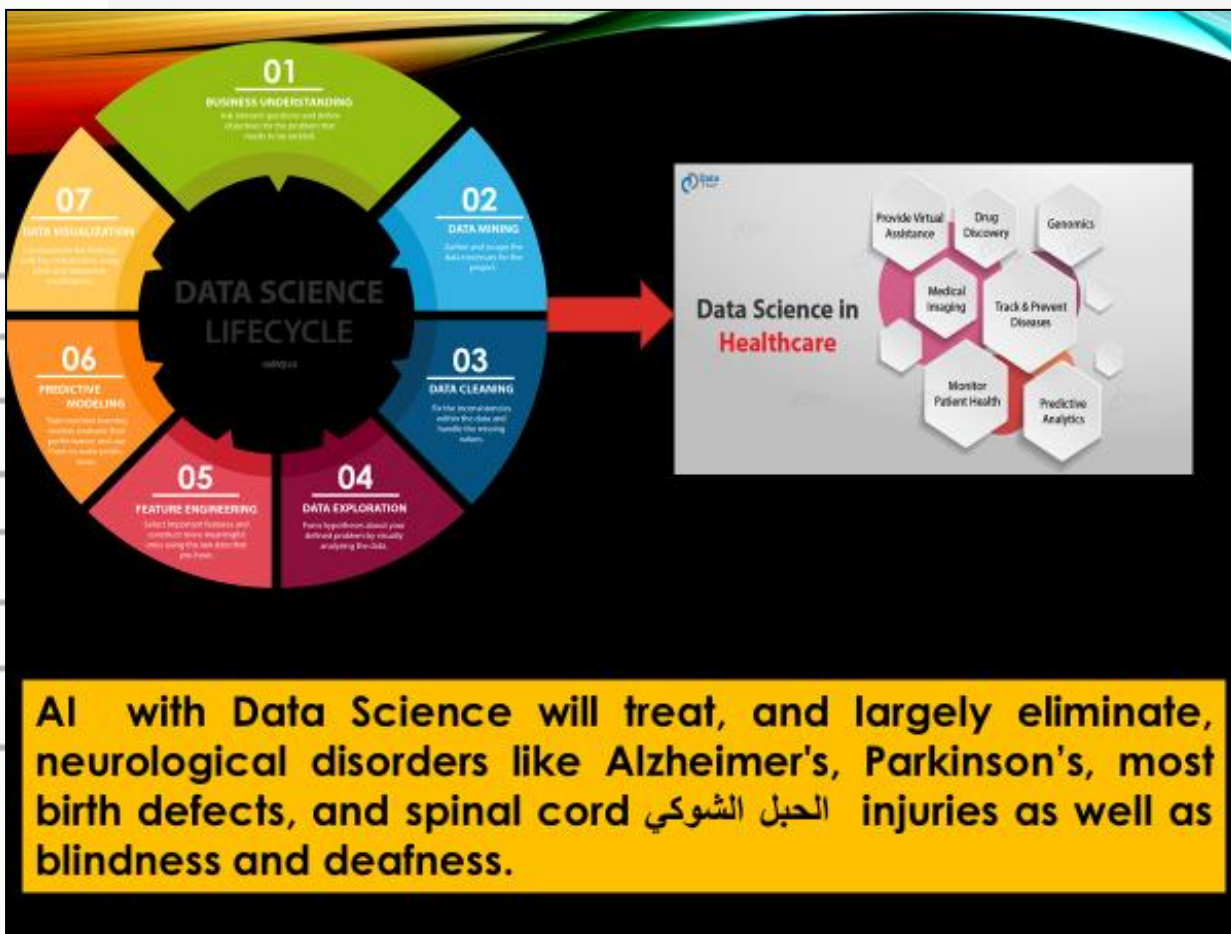
Midterm Model Answer

(a) Describe how the healthcare goes genomic and predictive in 2050.

5. HEALTHCARE GOES GENOMIC AND PREDICTIVE

- Prevention is better than cure.
- We're moving from a system of **generalized healthcare** based on some of your high-level identifiers like age, race, and Body Mass Index (BMI).
- We're moving to a world of **personalized medicine based on your sequence genome**.
- the **genome** (the entire set of DNA (Deoxyribonucleic acid) instructions found in a cell data will be used to innovate the prevention via the predictive analytics aimed at making a **prediction of future events** based on the available current data.
- Once you, and millions of others for algorithm training, are sequenced, then healthcare **will be predictive**.
- This will enable early intervention, prevention, and personalized treatment. That will improve the chances of recovery and survival.
- AI will also assist doctors in making accurate diagnoses to **identify patients' conditions early on, and avoid the risk of complications**.

Midterm Model Answer



Midterm Model Answer

(b) The Society of Automotive Engineers (SAE) defines six levels of vehicle driving automation systems. Explain these levels using the design of the future car shown in Fig.2

Different Cars Are Capable Of Different Levels Of Self-driving On A Scale Of 0-5 (6 levels)

- **Level 0:** All major systems are controlled by **humans**.
- **Level 1:** Certain systems, for example **cruise control** **مثبت السرعة** or **automatic braking**, can be controlled by the car.
- **Level 2:** At least two simultaneous automated functions such as **acceleration and steering** **توجيه** are offered by the vehicle. However, humans are required to ensure safe operation (emergency) .
- **Level 3:** The car is able to manage safety functions under certain conditions, but **the driver is expected to take over** when alerted.
- **Level 4:** The car is **fully-autonomous** in some driving scenarios, though not all.
- **Level 5:** The car is **completely autonomous** in every situation (2050).

Midterm Model Answer

(c) Describe a detailed daily life of a working lady on the first of January 2050.

OUR DAILY LIFE(2050)

Morning - AI Personal Assistants and Smart Homes

- **Wake-Up Routine:** The day begins with an **AI personal assistant** gently waking up Sarah, our hero, using a light and sound sequence optimized for her sleep cycle. The room's temperature and lighting adjust to her preferences, ensuring a comfortable day start.
- **Smart Wardrobe:** As Sarah gets ready, her **AI clothes** suggests an outfit based on the day's weather forecast and her schedule. The clothes are made of smart fabrics that adapt to temperature changes.
- **Nutrition and Health:** In the kitchen, her **AI nutritionist prepares a breakfast** customized to her eating needs, using ingredients from her smart fridge. Her **health monitoring system** updates her on her vitals and fitness progress.

Midterm Model Answer

OUR DAILY LIFE (2050)

Travel - Autonomous Vehicles and AI in Transit

- **Smart Travelling:** Sarah steps into her **self-driving car**, which communicates with **city traffic AI** to find the quickest, most efficient route. During the travel, she uses this time to **catch up on news**, selected by her **AI assistant** based on her interests.
- **Environmental Impact:** The car is **electric** and part of a shared vehicle network, significantly reducing carbon footprints and traffic crowding

Work - AI Collaboration and Enhanced Productivity

- **AI Coworkers:** At work, Sarah **collaborates with AI systems** that help with data analysis, creativity, and problem-solving, making her job more efficient and allowing her to focus on complex tasks.
- **Virtual Meetings:** Advanced **virtual reality meetings** allow real-time collaboration with colleagues worldwide, with AI translation removing any language barriers.

OUR DAILY LIFE (2050)

Afternoon - AI in Education and Leisure

- **Learning and Development:** Sarah attends an AI-powered education session, offering a personalized learning experience. The AI adapts the curriculum to her learning style and pace.
- **Leisure Activities:** For relaxation, she **visits a virtual reality park**, experiencing a safari adventure created by AI to simulate real-world experiences.

Evening - Social and Family Life

- **Home Cooking:** Back home, an **AI-powered kitchen** helps Sarah cook a meal, offering step-by-step instructions and adjusting recipes to her taste.
- **Family Time:** Sarah enjoys family time with interactive **AI-powered games** and educational tools, offering **a fun** and engaging way for **her children to learn and play**.

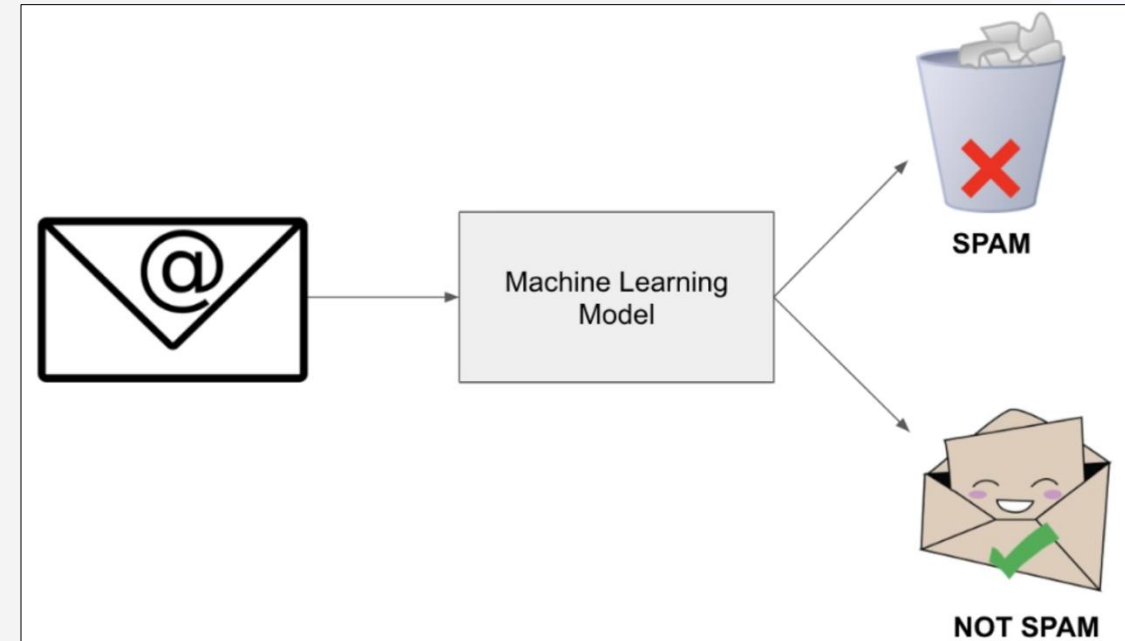
Night - Health and Comfort

- **Wellness Check:** Before bed, her health AI conducts a **quick wellness check**, adjusting her sleep environment to ensure a restful night.
- **Reflection and Planning:** Sarah reviews her day and **plans for tomorrow** with her AI assistant, which organizes her schedule and sets reminders for important tasks.

Classification Metrics

In a **classification task**, our main task is to **predict the target variable** which is in the form of **discrete values**. To evaluate the performance of such a model there are metrics as mentioned below:

1. Classification Accuracy
2. Confusion Matrix



1. Classification Accuracy

- **Accuracy** is the percentage of correct classifications that a trained machine learning model achieves,
- i.e., the **number of correct predictions divided** by the total number of predictions across **all classes**.



$$\text{Accuracy} = \frac{\text{Correct predictions}}{\text{All predictions}}$$

2. Confusion Matrix

- ✓ A **confusion matrix** is a table that is used to define the **performance of a classification algorithm**.
- ✓ A confusion matrix visualizes and **summarizes the performance** of a **classification algorithm**.

True Class			
		Positive	Negative
Predicated Class	Positive	TP	FP
	Negative	FN	TN

2. Confusion Matrix

There are 4 terms you should keep in mind:

- **True Positives:** It is the case where we predicted Yes and the real output was also yes.
- **True Negatives:** It is the case where we predicted No and the real output was also No.
- **False Positives:** It is the case where we predicted Yes but it was actually No.
- **False Negatives:** It is the case where we predicted No but it was actually Yes.

	Actually an Orange 106	Actually Not an Orange 60
Predicted Orange 115		
Predicted Not Orange 51		

Regression Evaluation Metrics

In the **regression task**, we are supposed to predict the target variable which is in the form of **continuous values**. To evaluate the performance of such a model below mentioned evaluation metrics are used:

1. Mean Absolute Error
2. Mean Squared Error
3. Root Mean Square Error



1. Mean Absolute Error

- ✓ It is the **average distance** between **Predicted and original** values.
- ✓ However, there is one limitation i.e. it doesn't give any idea about the **direction of the error** which is whether we are under-predicting or over-predicting our data.

$$MAE = \frac{1}{n} \sum \left| y - \hat{y} \right|$$

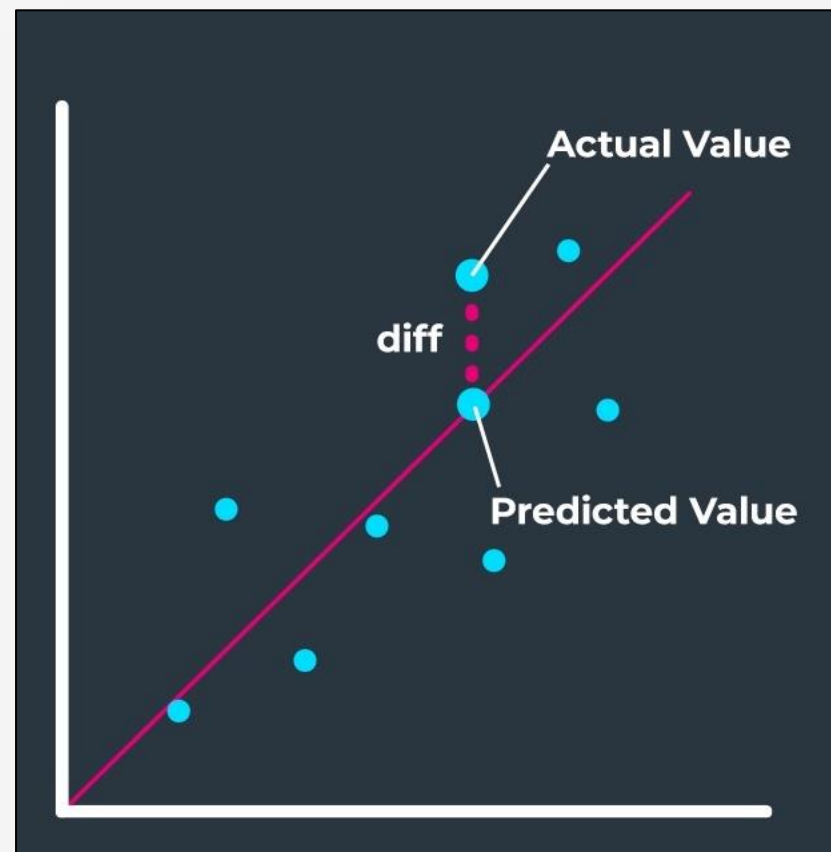
Divide by the total number of data points

Predicted output value

Actual output value

Sum of

The absolute value of the residual



2. Mean Squared Error

- ✓ It is **similar to mean absolute error** but the difference is it takes the **square of the average** of between **predicted** and **original** values.
- ✓ By taking the square of errors it pronounces larger errors more than smaller errors, we can **focus more on larger errors**.



$$\text{MSE} = \overset{\text{Mean}}{\frac{1}{n}} \sum_{i=1}^n \overset{\text{Error}}{(Y_i - \hat{Y}_i)} \overset{\text{Squared}}{^2}$$

3. Root Mean Squared Error

✓ **RMSE** is a metric that can be obtained by just taking the **square root of the MSE** value.

$$RMSE = \sqrt{\sum_{i=1}^n \frac{(\hat{y}_i - y_i)^2}{n}}$$

✓ RMSE is in the **same unit as the dependent variable**, making it more interpretable than MSE. It **combines the advantages of both MSE (sensitivity to large errors) and MAE (insensitivity to outliers)**.



Classification Model Evaluation – Accuracy in Python

```
from sklearn.model_selection import train_test_split
```

```
# Step 2: Split the data into training and test sets
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,
```

```
# Step 4: Train the model
```

```
knn = KNeighborsClassifier(n_neighbors=best_k)
```

```
knn.fit(X_train, y_train)
```

```
# Step 5: Make predictions
```

```
y_pred = knn.predict(X_test)
```

```
from sklearn.metrics import accuracy_score
```

```
# Step 6: Evaluate the model
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print(f'Accuracy: {accuracy}')
```

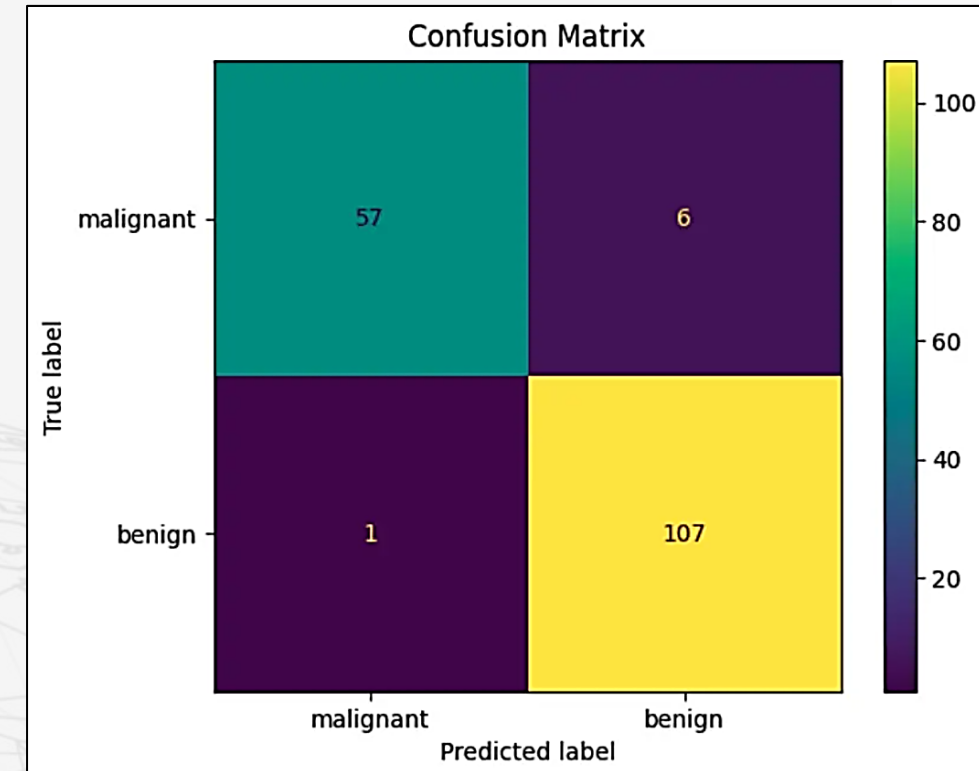
Accuracy: 0.9590643274853801

Classification Model Evaluation

Confusion Matrix in Python

```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

# Step 6: Plot the confusion matrix
cm = confusion_matrix(y_test, y_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=target_labels)
disp.plot()
disp.ax_.set(title='Confusion Matrix')
plt.show()
```



Regression Model Evaluation

Confusion Matrix in Python

```
import matplotlib.pyplot as plt
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.neighbors import KNeighborsRegressor
from sklearn.metrics import (
    mean_squared_error,
    mean_absolute_error,
    r2_score,
)

# Step 1: Load the dataset
data = load_diabetes()
X = data.data
y = data.target

# Step 2: Split the data into training and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random
```

```
# Step 3: Perform hyperparameter tuning with GridSearchCV
```

```
# Instantiate the model to be tuned
knn = KNeighborsRegressor()
```

Regression Model Evaluation

Confusion Matrix in Python

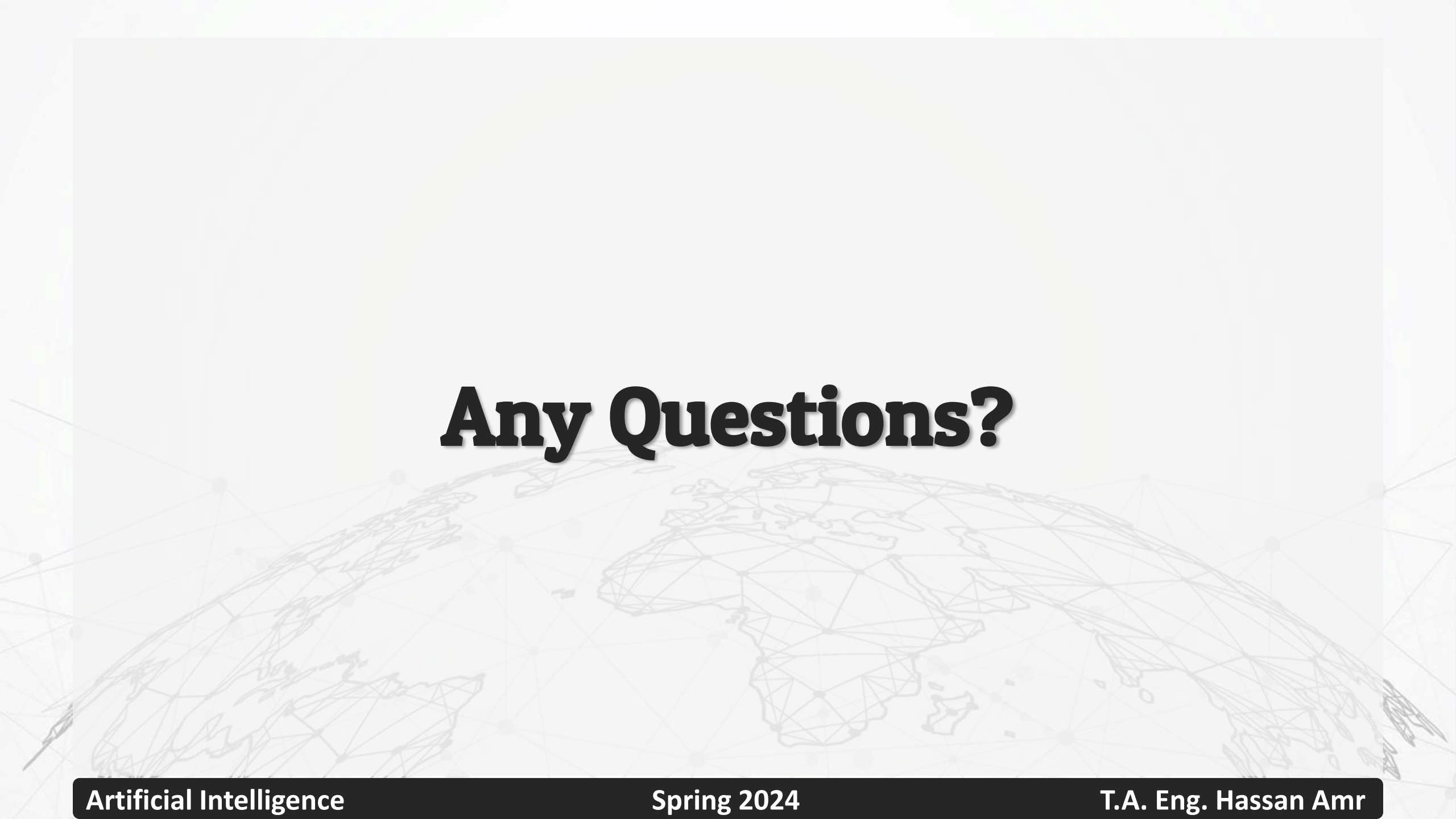
```
# Step 4: Train the model
knn = KNeighborsRegressor(n_neighbors=best_k)
knn.fit(X_train, y_train)

# Step 5: Make predictions
y_pred = knn.predict(X_test)

# Step 6: Evaluate the model
mse = mean_squared_error(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print(f'Mean Squared Error: {mse}')
print(f'Mean Absolute Error: {mae}')
print(f'R^2 Score: {r2}')
```

```
Mean Squared Error: 3175.5505430242274
Mean Absolute Error: 44.51629072681704
```

The background of the slide features a stylized, light gray globe. Overlaid on the globe is a complex network of thin gray lines connecting various points, resembling a global communication or data network. The text "Any Questions?" is centered in a large, bold, black font with a slight shadow effect.

Any Questions?